Suraj Kumar Kothuri

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Profile

PhD researcher in biophotonics at Tyndall National Institute, graduating early 2027, with four years of experience designing, building and validating compact optical systems. My expertise spans diffuse optics, light migration, tissue optics, time domain spectroscopy, free-space optical design, single-photon detection, miniaturisation and prototyping, photonics system validation on tissue phantoms and *in vivo*. Currently open to optical engineer or related positions.

Skills Summary

- **Optical systems**: Free-space optics, optical alignment, lasers, single-photon detectors, silicon photomultipliers, TCSPC, diffuse optics
- **Design and simulation**: Zemax, SolidWorks (3D), COMSOL Multiphysics, OptiFDTD
- **Clean Room**: Photonics packaging, Fiber array alignment, Wire bonding, Flip-chip bonding
- **Prototyping and integration**: Miniaturization, multimodal system integration, 3D printing, wire bonding
- **Test and validation**: System benchmarking on tissue phantoms (MEDPHOT), *in vivo* human studies, ex vivo studies
- **Software**: Python, MATLAB, data fitting
- **Courses & Training**: Photonics packaging, compound semiconductor device fabrication, micro transfer printing, Biophotonics

Soft Skills

- **Project Management**: Ability to manage multiple scientific projects at once.
- **Leadership Roles**: Mentored graduate and master's students for final year thesis.
- **Student representative**: Played a key role bridging communication between student communnity and management board.

Key Research Projects

- **Tyndall National Institute** — Cork, Ireland
  PhD Researcher, Biophotonics@Tyndall — *February 2023 – present*

  Wearable time-domain tissue oximetry
  - Developed a time domain NIRS system for tissue oxygenation monitoring from concept to prototype and system validation. The system is made from lidar components, its low cost, fast and suitable for wearable biomedical applications.
  - Established proof-of-concept through *in vivo* study by measuring arm-muscle oxygenation changes during and after cuff occlusion.

  Multimodal endoscopic probe
  - Integrated a time-of-flight channel into a 4.0 mm side-viewing endoscopic probe alongside a chip-on-tip micro-camera.
  - Benchmarked the probe on MEDPHOT phantoms for stability, repeatability and robustness. Coordinated phantom design with industry manufacturing partner BioPixS.

  Optical characterization of Human Bone
  - Led the first broadband (500–1100 nm) time-domain diffuse optical characterization of fresh human cadaver bone.
  - Published as first author in *Scientific Reports* with an open dataset.

Publications and Presentations

- "Broadband time domain diffuse optical characterization of human cadaver bone from 500 to 1100 nm." *Scientific Reports* (2025).
- "Multimodal endoscopy platform combining superficial tissue chip-on-tip micro-camera imaging and depth-resolved time-domain diffuse optical spectroscopy." *Journal of Physics: Photonics* (2026). Full list on Google Scholar.

Education

- **Tyndall National Institute (University College Cork)** — Cork, Ireland
  PhD — *2023 – 2027*

  Thesis: *Photonic Integration of Time-of-Flight Tools for Next-Generation Wearable Devices for Biomedical Applications*

- **Sri Sathya Sai Institute of Higher Learning** — India
  Integrated Masters in Physics (Photonics) — *2016 – 2022*